Type MS · Showing a cubic has only one real root

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

4 (a)	f (±1)=0 **or** f (±2)= –5	M1
	$-1 + 1p + -1q + 7 = 0$ $(p - q + 6 = 0)$ **and** $-8 + 4p + -2q + 7 = -5$ $(4p - 2q - 1 = -5)$	A1
	$4(q - 6) - 2q = -4$ $(2q = 20)$	dM1
	$p = 4$ $q = 10$	A1 A1 [5]
	ALT – polynomial division $(x^3 + px^2 + qx + 7) \div (x + 1) = x^2 + (p - 1)x + q - p + 1$ and comparison of final step of division with 7 to obtain an equation **or** $(x^3 + px^2 + qx + 7) \div (x + 2) =$ $x^2 + (p - 2)x + (q - 2p + 4)$ remainder -5 and comparison of final step of division with obtaining remainder -5 to obtain an equation	M1
	$(x^3 + px^2 + qx + 7) \div (x + 1) = x^2 + (p - 1)x + q - p + 1$ and comparison of final step of division with 7 to identify $q - p + 1 = 7$ **and** $(x^3 + px^2 + qx + 7) \div (x + 2) =$ $x^2 + (p - 2)x + (q - 2p + 4)$ remainder -5 and comparison of final step of division with obtaining remainder -5 to identify $7 - 2(q - 2p + 4) = -5$	A1
	$(p + 6) - 2p = 2$ $(-p = -4)$	dM1
	$p = 4$ $q = 10$	A1 A1
(b)	$x+1\overline{)x^3 + "4"x^2 + "10"x + 7}$ $(+3x + 7)$ x^2 $x^3 + x^2$ $"3"x^2$ or $x^3 + 4x^2 + 10x + 7 \equiv (x + 1)(x^2 + Ax + B)$	M1
	"3"^{2} – 4(1)("7") = a value	dM1
	–19 and a conclusion drawn e.g. the discriminant is negative so only one real root	A1 [3]
	ALT – use of completing the square $x+1\overline{)x^3 + "4"x^2 + "10"x + 7}$ $(+3x + 7)$ x^2 $x^3 + x^2$ $"3"x^2$ or $x^3 + 4x^2 + 10x + 7 \equiv (x + 1)(x^2 + Ax + B)$	M1

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	$\left(x+\frac{3}{2}\right)^2+\frac{19}{4}>0$	dM1
	$\left(x+\frac{3}{2}\right)^2+\frac{19}{4}>0$ and a conclusion drawn e.g. the completed square form is always greater than 0 so only one real root	A1
		Total 8 marks

Type MS · Showing a cubic has only one real root

Answer 1 of 5 · Question 4 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

	dM1	Attempt at completing the square See general guidance for what constitutes an attempt at completing the square
	A1	Correct completed square form and draws a conclusion. There must be some conclusion drawn, but it can be as simple as writing '#' or "shown".

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Answer 1 of 5 · Question 4 part b · 3 marks

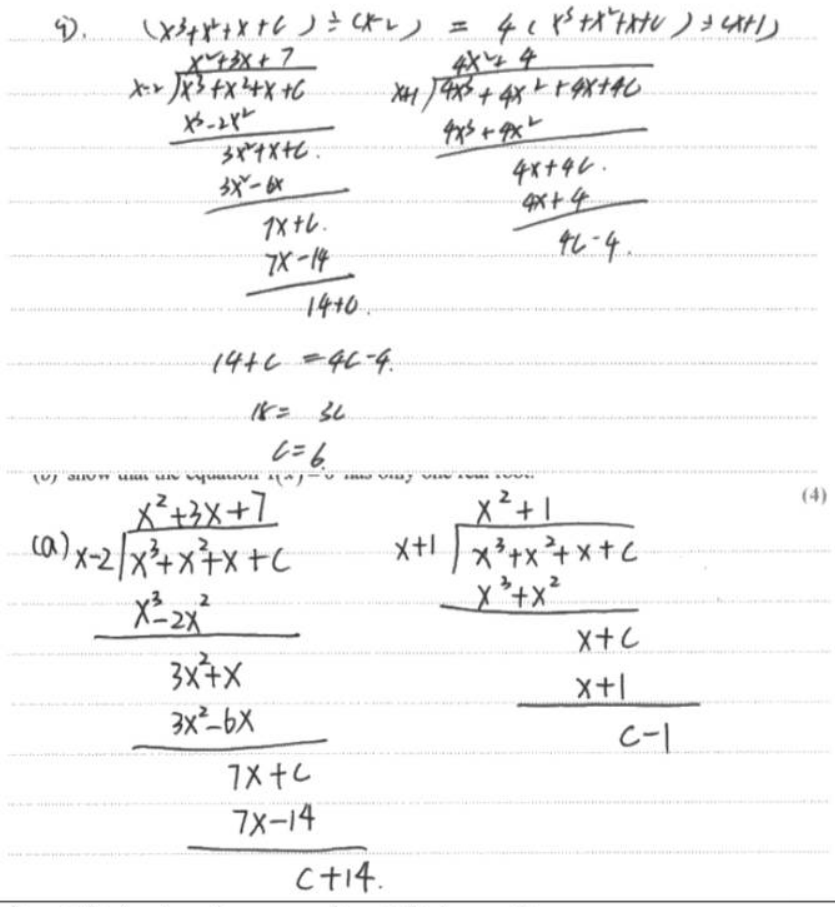
2022 JUN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Showing a cubic has only one real root

Answer 2 of 5 · Question 5 part b · 4 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

5 (a)	$4[(-1)^3+1^2+(-1)+c]=2^3+2^2+2+c$ $-4+4c=14+c$ oe $c=6$ *	M1 M1 ddM1 A1*cso [4]
		
(b)	$[(x+2)](x^2+Ax+3)$ or $[(x+2)](x^2-x+B)$ $[(x+2)](x^2-x+3)$ $(b^2-4ac=)(-1)^2-4(1)(3)<0$ therefore only one (real) root * or $\left(x-\frac{1}{2}\right)^2=-\frac{11}{4}$ and $-\frac{11}{4}<0$ therefore only one (real) root	M1 A1 M1 A1*cso [4]
		Total 8 marks

Type MS · Showing a cubic has only one real root

Answer 2 of 5 · Question 5 part b · 4 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correct substitution of – 1 **or** 2 into f(x)
	M1	For correct substitution of – 1 **and** 2 into f(x)
	For the first two marks: Where candidates employ simultaneous equations, mark the work for the substitution Accept eg -1 for $(-1)^3$ 4 for 2^2 Allow – 1 + c **with no errors** to imply substitution of – 1 Allow 14 + c **with no errors** to imply substitution of 2	
	ddM1	For $4[(-1)^3+1^2+(-1)+c]=2^3+2^2+2+c$ or condone the 4 on the 'wrong side'. $(-1)^3+1^2+(-1)+c=4[2^3+2^2+2+c]$ Dependent on both previous method marks. For forming simultaneous equations eg $4R = 14 + c$ $R = -1 + c$ Condone $R = 14 + c$ $4R = -1 + c$ The work doesn't have to be simplified for this mark.
	A1*cso	Correct solution, no errors or omissions, minimum steps as shown. The steps will need to be checked. Note there are different notations for long division.
	For a route that uses long division – examples in mark scheme. M1 For completing a correct long division, giving a correct remainder M1 for completing 2 correct long divisions, giving both correct remainders ddM1 for the correct equation or condone the 4 on the wrong side. A1 Correct solution, no errors or omissions, minimum steps as shown. The steps will need to be checked.	
(b)	M1	For $[(x+2)](x^2+Ax+3)$ or $[(x+2)](x^2-x+B)$ **using any method** – there must be a quadratic to award this mark. Mark the quadratic seen, not the method.
	A1	For x^2-x+3
	M1	For calculating the discriminant correctly for their 3 term quadratic. If working with the correct quadratic – 11 will imply this mark. If not, they must show at least partial substitution (their b^2 and $4ac$ evaluated will be fine) And condone/allow $(1)^2-4(1)(3)<0$ since $(-1)^2=(1)^2$ Or full correct completion of square and $-\frac{11}{4}<0$ oe. The discriminant may be calculated in the quadratic formula.
	A1*cso	Fully correct solution and short conclusion State < 0 and (can be "no (real) roots", (x =) "– 2 is the only (real) root", "shown" or #), no errors. Minimum steps as shown.
	Send any work on complex numbers to Review.	

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Answer 2 of 5 · Question 5 part b · 4 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Showing a cubic has only one real root

Answer 3 of 5 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

1. Factorisation:

$(x^2 + bx + c) = (x + p)(x + q)$,where $|pq| = |c|$ leading to $x = \ldots$

$(ax^2 + bx + c) = (mx + p)(nx + q)$ where$|pq| = |c|$ and $|mn| = |a|$leading to $x = \ldots$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x = \ldots$

3. Completing the square:

$x^2 + bx + c = 0$: $\left(x \pm \frac{b}{2}\right)^2 \pm q \pm c = 0, \quad q \neq 0$ leading to $x = \ldots$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $(x^n \rightarrow x^{n-1})$

2. Integration:

Power of at least one term increased by 1. $(x^n \rightarrow x^{n+1})$

Use of a formula:
Generally, the method mark is gained by **either** quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values **or**, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:
The rubric states "Without sufficient working, correct answers may be awarded no marks". General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

Type MS · Showing a cubic has only one real root

Answer 3 of 5 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks
1(a)	$f\left(\frac{2}{3}\right)=6\times\left(\frac{2}{3}\right)^3-13\times\left(\frac{2}{3}\right)^2+a\times\left(\frac{2}{3}\right)-10=0\Rightarrow\frac{2a}{3}=14$	M1
	$\Rightarrow a=21$*	A1 cso
	Alt – Polynomial division	[2]
	$3x-2\overline{)6x^3-13x^2+ax-10}\Rightarrow a-6=15\Rightarrow a=21$	[M1A1]
(b)	**Polynomial division**	
	$3x-2\overline{)6x^3-13x^2+21x-10}$	M1A1
	Equates coefficients	
	$(3x-2)(ax^2+bx+c)=6x^3-13x^2+21x-10$	
	$\Rightarrow 3a=6\Rightarrow a=2$	
	$\Rightarrow 3b-2a=-13\Rightarrow b=-3$	[M1
	$\Rightarrow -2c=-10\Rightarrow c=5$	
	Quadratic factor is $2x^2-3x+5$	A1]
	Conclusion	M1
	$b^2-4ac=(-3)^2-4\times2\times5=-31$	
	The discriminant is negative hence the quadratic has no real solutions and the only intersection with the x-axis is at $x=\frac{2}{3}$	A1 [4]
	ALT	
	Completes square:	
	$2x^2-3x+5=0\Rightarrow\left(x-\frac{3}{4}\right)^2=-\frac{31}{16}$	[M1
	$\left[\Rightarrow x=\frac{3}{4}\pm\sqrt{-\frac{31}{16}}\right]$	
	$-\frac{31}{16}<0$ so the quadratic has no real roots	
	The only intersection [root] with the x-axis is at $x=\frac{2}{3}$	A1]
		Total 6 marks

Type MS · Showing a cubic has only one real root

Answer 4 of 5 · Question 10 part d · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

10 (a)	$f\left(\frac{3}{4}\right)=16\left(\frac{3}{4}\right)^3+11\left(\frac{3}{4}\right)-15$ * $\left(\frac{27}{4}+\frac{33}{4}-15\right)=0$	M1 A1cso [2]
(b)(i)	$\left[\sin 2\theta=\sin(\theta+\theta)\right]=\sin\theta\cos\theta+\sin\theta\cos\theta$	M1
	$(\sin 2\theta=)2\sin\theta\cos\theta$ *	A1cso [2]
(b)(ii)	$\left[\cos 2\theta=\cos(\theta+\theta)=\right]\cos\theta\cos\theta-\sin\theta\sin\theta=\cos^2\theta-\sin^2\theta$	M1
	$=\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$ *	M1 A1cso [3]
(c)	$27\cos\theta(2\cos^2\theta-1)+19\sin\theta(2\sin\theta\cos\theta)-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38(1-\cos^2\theta)\cos\theta-15(=0)$	M1
	$54\cos^3\theta-27\cos\theta+38\cos\theta-38\cos^3\theta-15(=0)$ $(16\cos^3\theta+11\cos\theta-15=0)$ oe	A1
	(When $x=\cos\theta$) $16x^3+11x-15=0$ *	A1cso [4]
(d)	$(16x^3+11x-15=0\Rightarrow(4x-3)(4x^2+3x+5)=0)\rightarrow 4x^2+3x+5$	M1
	$b^2-4ac=9-80<0$ (So no real roots)	M1 A1
	(Only solution is) $4x-3=0$ or $4\cos\theta-3=0$ so $\cos\theta=\frac{3}{4}$ *	A1cso [4]
		Total 15 marks

Type MS · Showing a cubic has only one real root

Answer 4 of 5 · Question 10 part d · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Showing a cubic has only one real root

Answer 5 of 5 · Question 1 part b · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question		Working	Marks
1	(a)	$-3\times q=-12\Rightarrow q=4$	M1A1 [2]
	(b)	$b^2-4ac<0,\ \ (p-2)^2-4\times1\times'4'<0$ To find critical values; $p^2-4p-12=0\Rightarrow(p-6)(p+2)=0\Rightarrow p=6,\ -2$ (Chooses inside region from their critical values) $-2<p<6$	M1 dM1A1 ddM1 A1 [5]
	Alternative		
	(b)	$b^2-4ac<0,\ \ (p-2)^2-4\times1\times'4'<0$ $(p-2)^2<16\Rightarrow-4<(p-2)<4$ $-2<p<6$	M1 dM1A1 ddM1A1
			Total 7 marks

(a)	M1	Substitute $x=0$ and $f(x)=-12$ and proceed to eliminate the term in p to obtain a linear equation in q.
	A1	$q=4$ (score M1 A1 if no working shown)
(b)	For the first 3 marks allow $b^2-4ac<0$ or $b^2-4ac=0$ or $b^2-4ac>0$	
	M1	Attempt to use b^2-4ac (<0) with $b=p-2$, $a=1$ and $c=4$ (ft from (a)). Correct formula shown and used or correct substitution.
	dM1	Attempt to solve their quadratic inequality or equation. Depends on previous M mark. See general guidance for solving quadratic equations.
	A1	Correct critical values, $(p=)-2,(p=)6$
	ddM1	Correct use of their critical values, showing that the inside region has been selected. Both M marks needed. Allow with $\leq$ or $<$ or set language statement.
	A1	$-2<p<6$ must be in terms of p. Allow $p\in(-2,6)$ or $-2<p$ AND p < 6 or $\{-2<p\ \cap\ p<6\}$
Alternative		
(b)	For the first 3 marks allow $b^2-4ac<0$ or $b^2-4ac=0$ or $b^2-4ac>0$	
	M1	Attempt to use $b^2-4ac<0$ with $b=p-2$, $a=1$ and $c=4$ (ft from (a)). Correct formula shown and used or correct substitution.
	dM1	Write in the form $(p-2)^2<16$ and attempt to take square root of both sides (both positive and negative options are needed), or as $(p-2)^2-16<0$ and attempt to factorise $[(p-2)-4][[(p-2)+4]]<0$ Depends on first M mark.
	A1	Correct critical values, $p=-2$, $p=6$ (Accept if wrong variable is used.)
	ddM1	Correct use of their critical values, showing that the inside region has been selected. Both M marks needed. Allow with $\leq$ or $<$ or set language statement.
	A1	$-2<p<6$ (in terms of p) with options as above.

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Answer 5 of 5 · Question 1 part b · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks